SCHOOL OF APPLIED SCIENCE AND OPEN LEARNING

BACHELOR OF CYBER SECURITY

CS 340: Cybercrime Investigation and Forensic II

SESSIONAL EXAMINATION

DATE: **TIME: 3 hours**

INSTRUCTIONS

1. Write your Student Identification Number on the Answer Booklet provided
2. This paper has only 2 sections, Section A and Section B. [60 marks]
3. Section A contains open ended questions
4. Section B answer any 3 questions.
5. Please write as clearly as possible as illegible handwriting cannot be marked.
6. Number the answers to the questions clearly before answering

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Section A (Answer all Questions) [4 marks each question]

1. Which of the following is NOT a primary component of digital forensics ontology?
   a) Identification
   b) Preservation
   c) Exfiltration
   d) Analysis

2. What is file carving primarily used for in digital forensics?
   a) Malware analysis
   b) Reconstructing deleted or damaged files
   c) Encrypting files
   d) Network forensics

3. In memory forensics, which of the following is crucial for detecting malware?
   a) Static analysis
   b) Dynamic content analysis
   c) Signature-based detection
   d) Multi-media analysis

4. Open-source Intelligence (OSINT) in cybercrime investigations refers to:
   a) Hacking into secure databases
   b) Gathering publicly available information
   c) Decoding encrypted messages
   d) Extracting data from proprietary systems

5. Which technique is commonly used for fraud detection in forensic investigations?
   a) Data mining
   b) Social engineering
   c) File encryption
   d) Network segmentation

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SECTION B (Answer any 3 Questions) [40 marks]

QUESTION 1

During an investigation, you are given a **hard drive** suspected to contain malware. Explain the processes involved in conducting **malware forensics**. Discuss how both static and dynamic analysis can be used to uncover the behavior of the malicious code. **[13.3 marks]**

QUESTION 2

Scenario:

You are investigating a case where a suspect has attempted to delete files from their hard drive. However, you believe the files can be recovered using file carving techniques with a hex editor. The deleted file is a JPEG image that is partially corrupted but still contains identifiable header and footer information.

Task:

Describe the steps you would take to **recover** the deleted JPEG file using a **hex editor**. In your answer, include:

- The signature or markers that identify the start and end of a JPEG file.
- How you would locate these markers within a hex editor.
- The process of extracting the relevant hex data and saving it as a new file. **(7 Marks)**

I. Explain why file carving with a hex editor is useful in digital forensics, particularly when dealing with fragmented or partially corrupted files. **(3.15 Marks)**

II. Discuss any potential **limitations** or challenges of using file carving in this scenario, and how you might overcome them. **(3.15 Marks)**

[13.3 marks]

QUESTION 3

Scenario:

A hard drive has been formatted, and all visible files have been deleted. However, you suspect that important digital evidence still exists on the drive in the form of lost or deleted files. As a forensic investigator, you are tasked with recovering these files using Foremost, an open-source data recovery tool.

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Task:

Explain the steps you would follow to perform file recovery using Foremost. In your answer, include:

- How to specify the types of files you are looking to recover (e.g., JPEG, PDF, etc.).
- The command you would use to recover the files from a disk image.
- How the results of the recovery are structured and where the recovered files are stored. (7 Marks)

Discuss how Foremost is able to recover deleted files from a formatted disk. Explain the importance of file headers and footers in this process. (3.15 Marks)

Identify two possible challenges or limitations when using Foremost for data recovery, and suggest ways to address these challenges. (3.15 Marks)

[13.3 marks]

QUESTION 4

You are tasked with investigating a case of identity theft involving unauthorized access to a company's internal network. Outline the steps you would take to conduct a thorough digital forensic investigation. Include the following in your answer:

- Collection of evidence
- Tools used for analysis
- Steps to preserve the integrity of the evidence

[13.3 marks]

QUESTION 5

Scenario:

During a cybercrime investigation, you suspect that the suspect has used steganography to hide malicious data within an innocent-looking image file. Your task is to analyze the image and uncover any hidden information.

Task:

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I. Define steganography and explain how it differs from encryption. In your answer, describe how steganography can be used in cybercrime to conceal data and evade detection. (7 Marks)

II. Describe the steps you would take to detect and extract hidden data from an image file. Mention specific tools and techniques commonly used in forensic investigations for steganography analysis. (3.15 Marks)

III. Discuss two challenges investigators might face when trying to detect steganographic data in digital files, and suggest ways to overcome these challenges. (3.15 Marks)

[13.3 marks]

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